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Lab 03
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Camera Calibration

1 Estimating the Field of View

I think that my Google Pixel 9 has a field of view that is around 60° (on the normal zoom).

2 Manually Calculate the Field of View

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Measured Values:

- $wreal$ (length of the real object): 8.5 in.
- d (distance from the camera to the object): 6.25 in.
- wpx (length of the virtual image): 4032 px.

Focal Length:

$$f = d(wpx/wreal) \tag{1}$$

Field of View:

$$fov = 2 * arctan(w/2f) = 68.42 \tag{2}$$

3 OpenCV's Calculation of the Field of View

OpenCV estimated the field of view for my camera to be 3.07051048e+03. Using the same equation above, we would get the following.

$$fov = 2 * arctan(4032px/2 * 3,070.51048px) = 66.58 \tag{3}$$

Comparison

The manual calculation of the field of view was only off by around 2°. This is a 2.76% error.